Addition of angular momentum 1

For orbital angular momentum $\mathbf{L}$ we have

$$L_x\psi = -i\hbar\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)\psi$$

$$L_y\psi = -i\hbar\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}\right)\psi$$

$$L_z\psi = -i\hbar\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right)\psi$$

and for spin angular momentum $\mathbf{S}$

$$S_x\chi = \frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\chi$$

$$S_y\chi = \frac{\hbar}{2}\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}\chi$$

$$S_z\chi = \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\chi$$

Let $\mathbf{J}$ be total angular momentum

$$\mathbf{J} = \mathbf{L} + \mathbf{S}$$

Let Ψ be the product of wave function ψ and spin vector χ.

$$\Psi = \psi\chi$$

Then

$$\mathbf{J}\Psi = \mathbf{L}\Psi + \mathbf{S}\Psi$$

Let J^2 be the magnitude-squared of total angular momentum.

$$J^2 = \mathbf{J}\cdot\mathbf{J} = J_x^2 + J_y^2 + J_z^2$$

Operator J^2 can be decomposed as

$$J^2 = (\mathbf{L} + \mathbf{S})\cdot(\mathbf{L} + \mathbf{S}) = L^2 + S^2 + 2\mathbf{L}\cdot\mathbf{S} \tag{1}$$

Verify J^2 commutes with L^2 and S^2.

$$[J^2, L^2] = [J^2, S^2] = 0$$

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